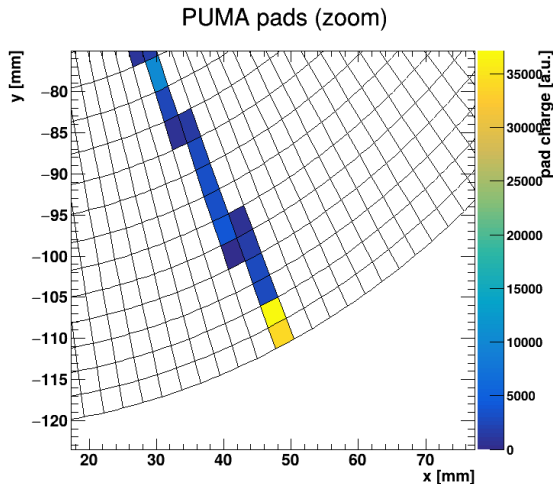


PUMA TPC — track reconstruction

Event representation, pulse analysis, tracking

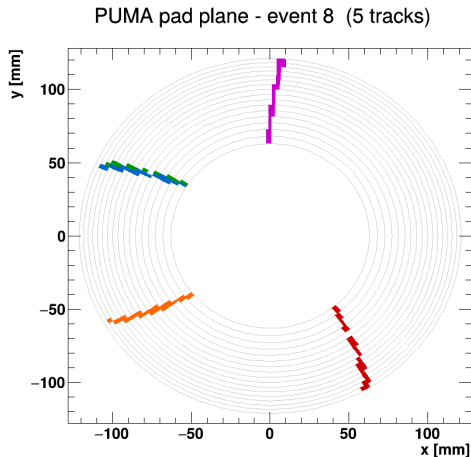
The PUMA active volume

- Annular gas TPC: **P10 @ 1 bar**, $R = 62.9\text{--}121.1\text{ mm}$, drift length 300 mm , **$B = 4\text{ T}$** solenoid.
- Pad plane: **16 equal-area rings** $\times$ **256 azimuthal pads** = 4096 pads.
- A charged track ionises the gas $\rightarrow$ electrons drift to the pads.
- Each pad gives (x, y) from its position and z from the **drift time** of the pulse.



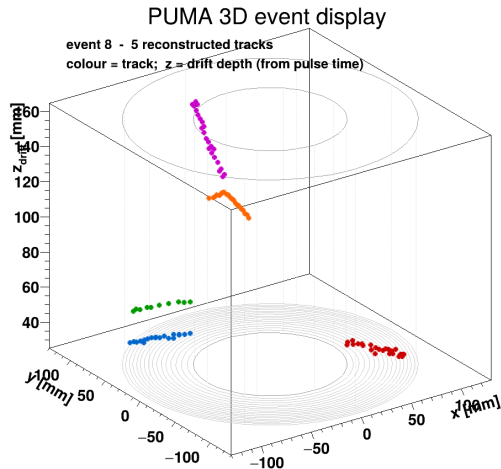
A track fires a column of pads; colour = charge.

A reconstructed event — 2D pad plane



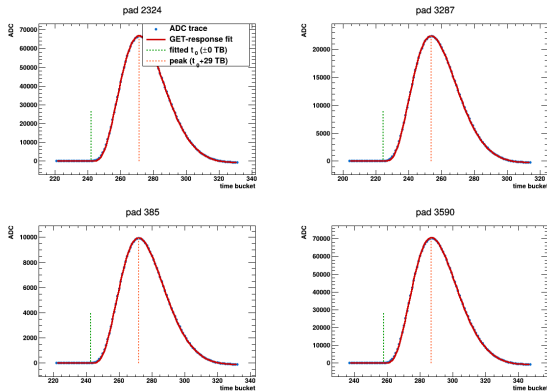
Each fired pad drawn as its real annular sector, coloured by the pattern-recognition track it belongs to.
Five tracks; the concentric circles are the 16 ring boundaries.

The same event — in 3D



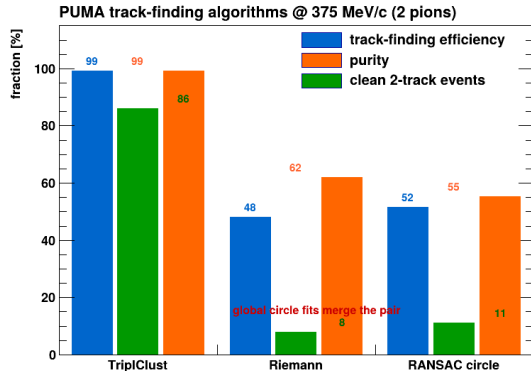
Same five tracks in (x, y, z_{drift}) : the pad gives (x, y) , the pulse arrival time gives z . Tracks sit at different drift depths inside the annular TPC volume.

From pad pulses to hits — the PSA fit



Pad ADC traces (blue) with the **GET electronics-response fit** (red, AtPSAMultiFit: $A e^{-3u} \sin u u^3$, $u = (t - t_0)/\tau$, $\tau = 25$ TB) — overlays the pulse shape **perfectly**. The fitted **impulse time** t_0 gives the **unbiased** drift depth (z); the **response peak** lags t_0 by the ~ 8.6 mm shaping delay. One fitted pulse $\rightarrow$ one AtHit.

Finding tracks: cluster first, then fit



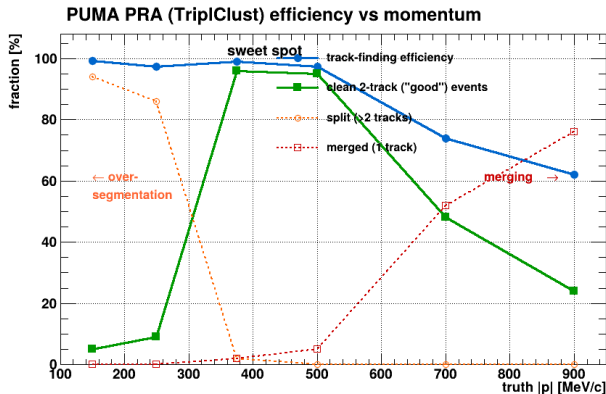
Pattern recognition groups the AtHits into tracks. Two paradigms:

- **spatial clustering** — TripClust, HDBSCAN
- **global circle consensus** — Riemann, RANSAC

On the 2-pion channel at 375 MeV/c, **clustering finds both tracks** (99% eff, 99% purity, 86% clean events). The global circle fits fit *one* dominant circle and **merge the back-to-back pair** (~50% eff, ~10% clean).

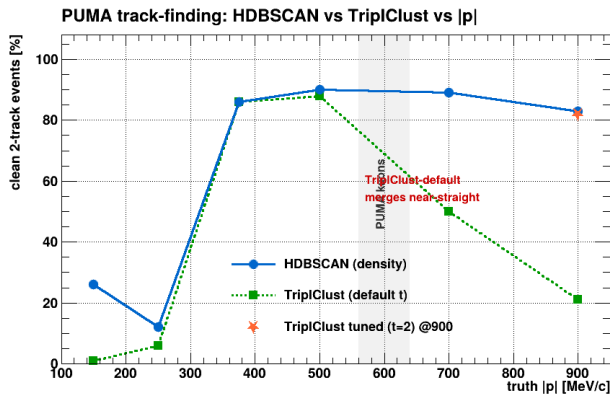
Lesson: cluster the hits first, then fit each track — don't fit one global circle to a multi-track event.

Track-finding efficiency vs momentum



Truth-matched efficiency & clean-2-track fraction (TriplClust). **Sweet spot 375–500 MeV/c** (95% clean — the PUMA pion range). Two failure modes: **over-segmentation** at low p (tight curls fragment into >2 tracks) and **merging** at high p (near-straight tracks fuse). Efficiency stays high; *event quality* degrades at both ends.

HDBSCAN vs TriplClust — robustness at high p



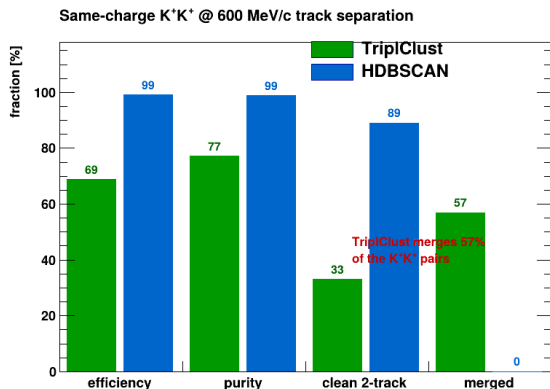
Same reco, two clustering finders:

- **HDBSCAN** (density) holds **83–90%** clean events from 375–900 MeV/c — it does *not* merge near-straight tracks.
- **TriplClust** (default t) matches it in the sweet spot but **collapses above 500 MeV/c**.

Crucial for the **PUMA kaons** (~ 600 MeV/c, high- p). TriplClust *can* recover with a lowered cluster distance ($t = 2$ @900 $\rightarrow 82\%$, $\star$) — but needs momentum-adaptive tuning.

Recommendation: HDBSCAN as the default track finder (or adaptive- t TriplClust).

The decisive case: same-charge K^+K^+

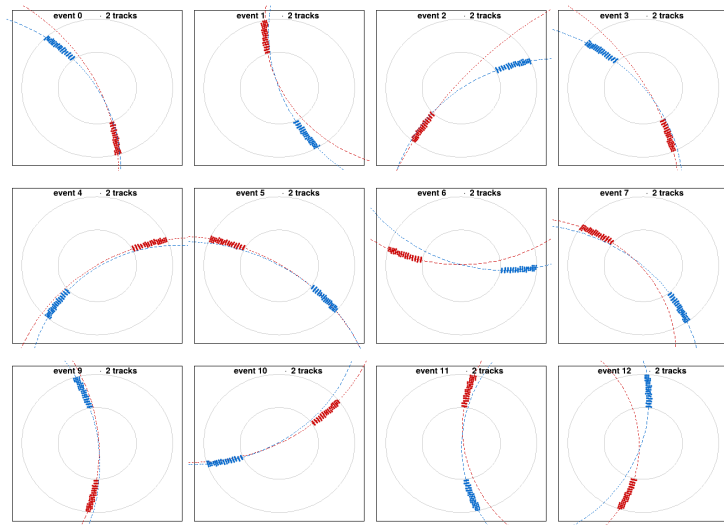


The real PUMA target: $^3\text{He} + \bar{p} \rightarrow K^+K^+\pi^-$. The two kaons are **same charge** (mirror circles) and **near-straight** at 600 MeV/c — the worst case for track separation.

TripClust merges 57% of the pairs (33% clean events). **HDBSCAN separates them cleanly** — 99% efficiency, 99% purity, **89% clean events, 0% merged**.

This is the case that decides the track finder: **HDBSCAN for PUMA**.

Pattern recognition — examples



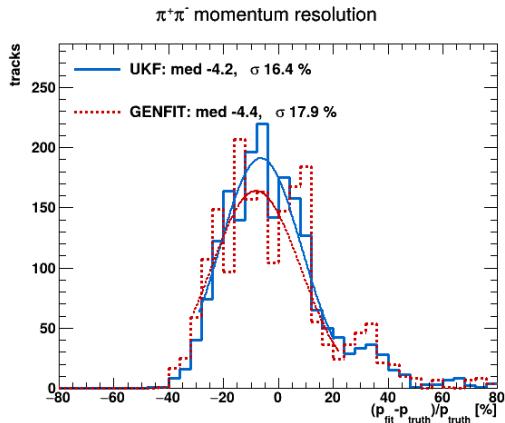
Twelve events: hits coloured per track candidate, with the PRA circle fit (dashed) overlaid. The DLC dispersed charge gives the fattened hit bands

Two Kalman fitters

Each reconstructed track is fit by **two** independent Kalman filters on the *same* clusters:

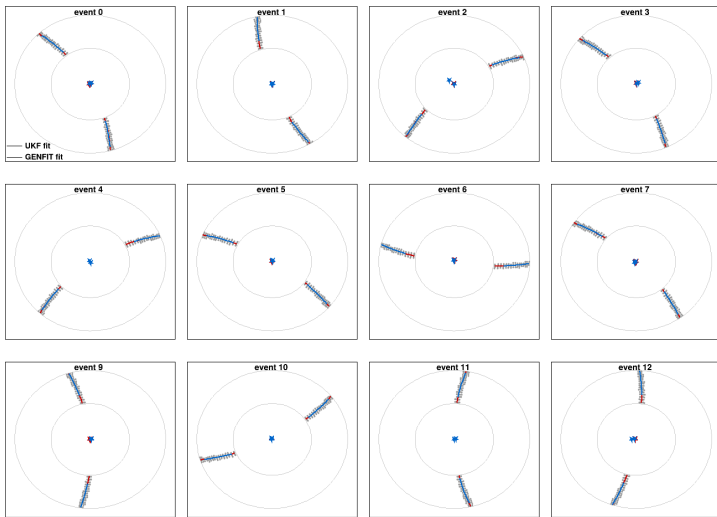
- **UKF** — AtFitterUKF: unscented Kalman filter + CATIMA energy loss (PUMA-tuned).
- **GENFIT** — KalmanFitterRefTrack: reference-track KF, native Bethe-Bloch.

Each fitted track yields momentum, polar angle, charge sign, and the back-extrapolated vertex.



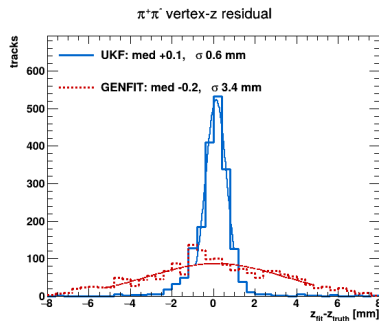
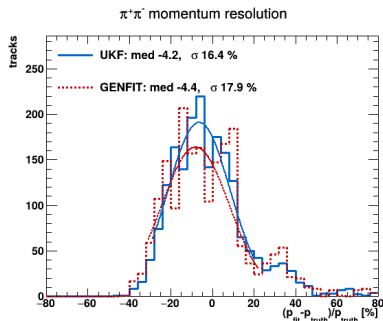
$\pi^+\pi^-$ momentum residual: both fitters, $\sigma \sim 16\text{--}18\%$.

Kalman fits — examples



Twelve events: reconstructed hits (grey) with the UKF and GENFIT fitted trajectories overlaid; the stars are the back extrapolated vertices (converging on the beam axis at the centre)

Momentum, angle & vertex resolution

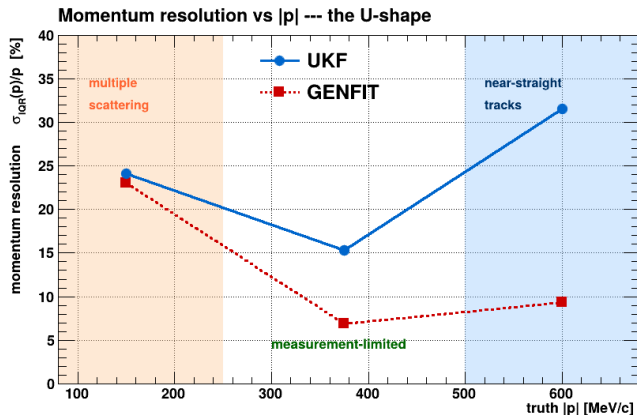


$\pi^+\pi^-$, 1000 events, identical clusters.

	tracking	fitting	momentum σ	θ σ	vertex σ_z
UKF	99.6%	99.4%	16%	2.1°	0.62 mm
GENFIT	99.6%	99.4%	18%	4.9°	3.4 mm

charge-sign **99.5%** (both); tracking + fitting efficiency **~99.5%** at 375 MeV/c. UKF wins on angle & vertex.

Resolution vs momentum — the U-shape

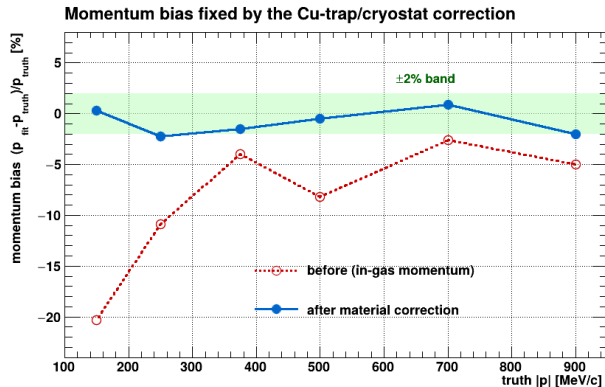


The momentum resolution is best in the middle (375–500 MeV/c) and degrades at both ends:

- **Low p : multiple scattering** ($\propto 1/p\beta$) perturbs the curvature — an *irreducible* floor. Both fitters converge, and tight curls also over-segment.
- **High p : tracks are near-straight**, with little curvature/sagitta to measure.

Only in the measurement-limited middle does the sub-pad information help — where GENFIT pulls ahead.

The momentum bias — and its fix

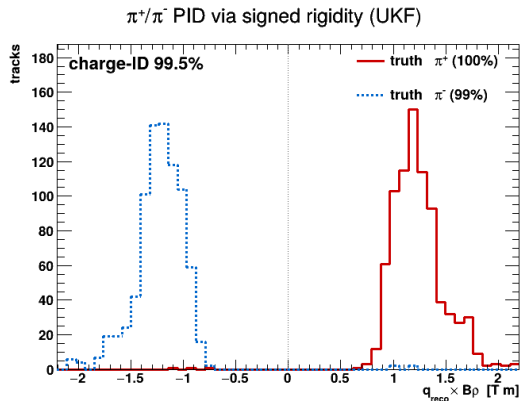


Momentum came out biased *low* (-4% @375, -20% @150 MeV/c). **Not a reco error** — the reco measures the in-gas momentum correctly.

The annihilation vertex sits inside a **Cu antiproton trap + Al cryostat**; the pion loses ~ 16 MeV escaping to the gas *before* it is tracked (Bethe-Bloch, worse at low p).

Fix: a deterministic vertex material correction (both fitters) $\rightarrow$ bias **within** $\pm 2\%$ across 150–900 MeV/c.

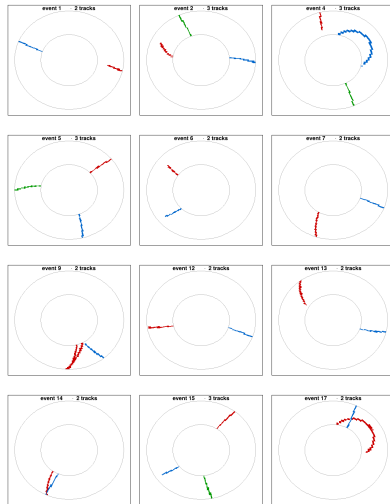
Particle ID: π^+ / π^- separation



π^+ and π^- have **identical mass** $\rightarrow$ identical dE/dx . They separate purely by **charge sign** — the curvature sense in the 4 T field, i.e. signed rigidity $q B\rho = \text{sign} \cdot p/0.2998$ [T m].

Truth-labelled, 1000 events: charge-ID **99.5%**, separation power **$S = 5.9 \sigma$** . The $\sim 0.5\%$ cross-over is the mis-ID.

Gallery of reconstructed events



Each fired pad as its annular sector, coloured by reconstructed track. 2–3 well-separated tracks per event, various curvatures (momenta).